A&A, 713, A226 (2026)
https://doi.org/10.1051/0004-6361/202659127
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Astronomy & Astrophysics

Post-Newtonian theory for precise relativistic astrometry

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Received 26 January 2026 / Accepted 10 June 2026

ABSTRACT

Context. High-precision astrometric observations at the microarcsecond level require a consistent relativistic framework for modelling both the dynamics of test particles and the propagation of light in gravitational fields.
Aims. We aim to formulate the post-Newtonian (PN) expansion in the test body limit in a manner that yields simple and broadly applicable expressions for the motion of massive and massless particles in generic, spherically symmetric spacetimes. In particular, we intend to formulate it such that the formalism suits the modelling of observations of stars orbiting the Galactic centre (GC).
Methods. We employed the PN expansion scheme to derive equations of motion for test particles and applied a perturbative solution method to solve the boundary value problem of identifying the geodesic connecting a given emitter–observer pair. Results are validated against numerical integration of the photon geodesic equation.
Results. We derive the PN equations of motion for test particles in a generic, spherically symmetric spacetime. We derive analytical expressions for the apparent position of stars as measured by an observer on Earth and show the importance of this formalism for the GC science case. We compare our results with other approaches taken in the literature and discuss its implications for the modelling of orbital motion around Sgr A*.
Conclusions. The proposed framework is well suited for the analysis of astrometric observations of stars orbiting the GC with microarcsecond accuracy. In this context, measurements performed with the GRAVITY+ instrument at the Very Large Telescope Interferometer can be used to place constraints on metric theories of gravity through the combined modelling of stellar dynamics and relativistic light propagation near Sgr A*.

Key words. black hole physics – gravitation – methods: analytical – astrometry – celestial mechanics – galaxy: center

1. Introduction

The post-Newtonian (PN) formalism has shown to be an extremely powerful and effective mathematical framework with which to explore gravitational phenomena (Will 2011). It has played a crucial role in interpreting experimental tests of general relativity (GR; Will 1993) and is today a key tool for studying the gravitational wave emission from compact binaries (Blanchet et al. 2010; Abbott et al. 2017; Marchand et al. 2020) and the environmental effects that shape the evolution of such astrophysical systems (Cardoso & Maselli 2020). Within the Solar System, the PN formalism is key for precise celestial mechanics (Kopeikin et al. 2011), and its applications go beyond astrophysics, with global navigation satellite systems (GNSS; Teunissen & Montenbruck 2021) being the most notable of its uses.

The PN formalism is an expansion scheme used to study Einstein's field equations in the small-velocity and weak-field regime. For a gravitational system with typical velocity, v; mass, M; and size, R, this regime is characterised by the expansion parameter

$$\epsilon \sim \left(\frac{v}{c}\right)^2 \sim \left(\frac{GM}{c^2 R}\right), \tag{1}$$

where G and c are Newton's gravitational constant and the speed of light, respectively. The strong gravitational regime is characterised by $\epsilon \sim 1$, while Newtonian gravity is retrieved in the limit $\epsilon \to 0$.

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In the Solar System, where $\epsilon \sim 10^{-8}$, the observed light deflection (Fomalont et al. 2009; Bruns 2018), time delay (Shapiro et al. 2004) and gravitational redshift (Lopresto et al. 1991) agree with the GR predictions to 1PN order. Outside the Solar System, 2.5 PN order effects have been observed in the orbital motion of the Hulse–Taylor binary pulsar (Taylor et al. 1979) and more recently in the double pulsar PSR J0737–3039 (Kramer & Wex 2009; Kramer et al. 2021).

In the Galactic centre (GC) where stars orbiting Sagittarius A* (Sgr A*) can reach orbital velocities with $\epsilon \sim 10^{-3}$, similar agreement is found with the predicted gravitational redshift (GRAVITY Collaboration 2019, 2018; Do et al. 2019) and orbital motion (GRAVITY Collaboration 2020) to 1PN order. The feasibility of such measurements allows deviations from GR and alternatives to the black-hole (BH) hypothesis to be experimentally tested using the bright stars in the S-cluster surrounding Sgr A*. The recent works by the GRAVITY Collaboration constrain deviations from GR due to the presence of extra fields (GRAVITY Collaboration 2023) or Yukawa-like forces (GRAVITY Collaboration 2025). These results rely on the precise modelling of both the orbital motion of stars as well as their appearance to a distant observer on Earth. Relativistic effects such as the transverse Doppler shift or the gravitational redshift are routinely taken into account. Others, like the role of light bending, are typically not addressed since these effects fall below the precision of the observations. With the advent of GRAVITY+ (GRAVITY+ Collaboration 2022) and its increased astrometric precision, a consistent formalism is required to model these observations.

In this manuscript, we outline and develop a formalism suited for self-consistent and robust analysis of GC observations with GRAVITY+. The paper is organised as follows. In Sect. 2, we derive the generic form of the equations of motion for massive and massless particles in spherically symmetric spacetimes. Sections 3 and 4 are dedicated to the generic problem of calculating observables in general relativity. In Sect. 3, we solve the boundary-value problem for geodesics connecting any two points in spacetime. In Sect. 4, we derive the redshift and time-delay expressions in spherically symmetric spacetimes. Section 5 showcases the applicability of the framework to the GC science case. We end with a discussion of our results and contextualise them with other approaches in the literature. Throughout the paper, Greek indices run from 0 to 3 and Latin ones from 1 to 3. We refrain from using geometric units in the derived expressions.

2. The expansion method

Spherically symmetric spacetimes provide a natural setting to study a wide class of gravity theories. Besides GR, scalar-tensor theories such as Brans–Dicke (Brans & Dicke 1961) or Horndeski gravity (Horndeski 1974; Kobayashi 2019), as well as $f(R)$ modifications of GR (Sotiriou & Faraoni 2010), admit spherically symmetric solutions in their field equations. Symmetry restricts the metric to depend on a single coordinate, r, allowing the line element to be written in spherical coordinates as

$$ds^2 = -p(r)\,c^2dt^2 + q(r)\,dr^2 + a(r)^2\,d\Omega^2, \tag{2}$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta\,d\phi^2$ is the line element on the two-sphere. The functions $p(r)$, $q(r)$, and $a(r)$ are not fixed by symmetry alone but are instead determined by the field equations of each theory. This allows us to work in a theory-agnostic framework. The Cartesian form of this line element can be obtained by defining a coordinate system (t, x^1, x^2, x^3) such that the spatial components relate according to the usual spherical coordinate transformations. Defining the radial coordinate as $r = \sqrt{\delta_{ij}x^ix^j}$ and the Euclidean-like unit vectors as $n_i = \delta_{ij}x^j/r$, the resulting metric reads

$$ds^2 = -A(r)c^2dt^2 + \Big[B(r)\delta_{ij} + D(r)\,n_in_j\Big]\,dx^idx^j, \tag{3}$$

where the metric coefficients relate to the spherical ones as

$$A(r) = p(r), \quad B(r) = \frac{a(r)^2}{r^2}, \quad D(r) = q(r) - \frac{a(r)^2}{r^2}. \tag{4}$$

2.1. Equations of motion for massive particles

To study the motion of test particles we write the geodesic equation explicitly as a function of the time coordinate t,

$$\frac{d^2x^k}{dt^2} = -\left[\Gamma^k_{\sigma\lambda} - \Gamma^0_{\sigma\lambda}\left(\frac{v^k}{c}\right)\right]v^\sigma v^\lambda, \tag{5}$$

where $\Gamma^\mu_{\sigma\lambda}$ are the Christoffel symbols and $v^k = dx^k/dt$ represents the velocity of the particle. Throughout the manuscript we denote the spatial components of the particle's position and velocity by $\boldsymbol{x}$ and $\boldsymbol{v}$, respectively. Evaluating the Christoffel symbols using the Cartesian metric coefficients yields the exact expression for the Newtonian-like equations of motion,

$$\frac{d^2\boldsymbol{x}}{dt^2} = \Big[c^2\,\mathcal{T}(r) + v^2\,\mathcal{V}(r) + (\boldsymbol{n}\cdot\boldsymbol{v})^2\mathcal{N}(r)\Big]\,\boldsymbol{n} + (\boldsymbol{n}\cdot\boldsymbol{v})\,\mathcal{H}(r)\,\boldsymbol{v}, \tag{6}$$

where $\boldsymbol{n} = \boldsymbol{x}/r$ is the radial direction and the pre-factor functions are given by

$$\mathcal{T}(r) = -\frac{A'}{2(B+D)}, \quad \mathcal{V}(r) = \frac{1}{2(B+D)}\left(B' - \frac{2D}{r}\right), \quad \mathcal{N}(r) = -\frac{1}{2(B+D)}\left(D' - \frac{2D}{r} + \frac{2DB'}{B}\right), \quad \mathcal{H}(r) = \left(\frac{A'}{A} - \frac{B'}{B}\right), \tag{7}$$

with the prime notation denoting derivatives with respect to the radial coordinate. The expansion of these functions in powers of ϵ yields the PN acceleration to the corresponding expansion order:

$$\frac{d^2\boldsymbol{x}}{dt^2} = \boldsymbol{a}_{\rm Newton} + \boldsymbol{a}_{\rm 1PN} + \boldsymbol{a}_{\rm 2PN} + \boldsymbol{a}_{\rm 3PN} + \cdots, \tag{8}$$

where the first term is the usual Newtonian acceleration

$$\boldsymbol{a}_{\rm Newton} = -\frac{GM}{r^2}\boldsymbol{n}, \tag{9}$$

and the remaining terms represent higher order corrections. Since the functions defined in (7) have units of inverse distance, the most natural expansion convention to use is

$$\mathcal{F}(r) = \sum_{n=1}^{\infty} \mathcal{F}_n \left(\frac{1}{r}\right)\left(\frac{GM}{c^2 r}\right)^n, \tag{10}$$

where $\mathcal{F}(r)$ represents any of the functions in (7). The convention leads to the generic form of each PN acceleration term:

$$\boldsymbol{a}_{n\rm PN} = \frac{1}{r}\left(\frac{GM}{c^2 r}\right)^n \left\{ \left(\mathcal{T}_{n+1}\frac{GM}{r} + v^2\,\mathcal{V}_n + (\boldsymbol{n}\cdot\boldsymbol{v})^2\mathcal{N}_n\right)\boldsymbol{n} + (\boldsymbol{n}\cdot\boldsymbol{v})\,\mathcal{H}_n\,\boldsymbol{v}\right\}. \tag{11}$$

The Cartesian form of the metric (3) naturally separates the equation of motion into components along the radial and velocity directions with relatively simple pre-factor functions. Furthermore, because the equation of motion describes massive particles, every expansion term, $\boldsymbol{a}_{n\rm PN}$, contains a 'Newtonian-like' acceleration term associated with the $\mathcal{T}_{n+1}$ coefficient. From the expansion convention we see that, for asymptotically flat spacetimes, the zeroth-order coefficients must be

$$\mathcal{T}_0 = \mathcal{V}_0 = \mathcal{N}_0 = \mathcal{H}_0 = 0. \tag{12}$$

Furthermore, for a physically sensible metric solution, $\mathcal{T}_1 = -1$, so that the zeroth-order acceleration matches that expected from the Newtonian limit.

2.2. Equations of motion for massless particles

A similar procedure can be conducted for massless particles with two caveats. First, because the particle must follow a null geodesic, the particle's four-velocity must satisfy the following condition:

$$g_{\mu\nu}\frac{dx^\mu}{dt}\frac{dx^\nu}{dt} = 0. \tag{13}$$

Second, precisely because of the null condition above, one must take into account that the three-velocity norm of a photon is c. In this case, we write the geodesic equation in terms of $\bar{t} = ct$ as

$$\frac{d^2x^k}{d\bar{t}^2} = -\left[\Gamma^k_{\sigma\rho} - \Gamma^0_{\sigma\rho}\left(\frac{dx^k}{d\bar{t}}\right)\right]\left(\frac{dx^\sigma}{d\bar{t}}\right)\left(\frac{dx^\rho}{d\bar{t}}\right) \tag{14}$$

and follow the same steps as before. The equation of motion then reads

$$\frac{d^2\boldsymbol{x}}{d\bar{t}^2} = \left[\mathcal{T}(r) + \left|\frac{d\boldsymbol{x}}{d\bar{t}}\right|^2 \mathcal{V}(r) + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)^2 \mathcal{N}(r)\right]\frac{\boldsymbol{x}}{|\boldsymbol{x}|} + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)\mathcal{H}(r)\,\frac{d\boldsymbol{x}}{d\bar{t}}, \tag{15}$$

and the expansion naturally follows:

$$\left(\frac{d^2\boldsymbol{x}}{d\bar{t}^2}\right)_{nPN} = \frac{1}{r}\left(\frac{GM}{c^2 r}\right)^n \left\{\left(\mathcal{T}_n + \left|\frac{d\boldsymbol{x}}{d\bar{t}}\right|^2 \mathcal{V}_n + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)^2 \mathcal{N}_n\right)\frac{\boldsymbol{x}}{|\boldsymbol{x}|} + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)\mathcal{H}_n\frac{d\boldsymbol{x}}{d\bar{t}}\right\}. \tag{16}$$

Notice that, unlike in Eq. (11), no mixed-order coefficients are present. The null vector condition (13) can similarly be expanded. When written in the Cartesian form, it reads

$$\mathcal{A}(r) - \mathcal{B}(r)\left|\frac{d\boldsymbol{x}}{d\bar{t}}\right|^2 + \mathcal{D}(r)\left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)^2 = 0. \tag{17}$$

Contrary to the equations of motion, it depends directly on the metric functions and not their derivatives. Hence, as opposed to the expansion form in Eq. (10), these functions are expanded according to

$$\mathcal{F}(r) = \sum_{n=0}^{\infty} \mathcal{F}_n \left(\frac{GM}{c^2 r}\right)^n, \tag{18}$$

where $\mathcal{F}(r)$ represents $\mathcal{A}(r)$, $\mathcal{B}(r)$, or $\mathcal{D}(r)$. This convention leads to the series expansion

$$\mathcal{A}_n + \mathcal{B}_n \left|\frac{d\boldsymbol{x}}{d\bar{t}}\right|^2 + \mathcal{D}_n \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|} \cdot \frac{d\boldsymbol{x}}{d\bar{t}}\right)^2 = 0, \tag{19}$$

which must be consistently evaluated at the same order as the equations of motion. Because this condition depends directly on the metric functions, the zeroth-order expansion coefficients are not zero. For asymptotically flat spacetimes, however, the time and radial coordinates can always be defined such that these coefficients match those associated with the Minkowski metric:

$$\mathcal{A}_0 = -1 \quad \mathcal{B}_0 = 1 \quad \mathcal{D}_0 = 0. \tag{20}$$

2.3. Schwarzschild coefficients

The Schwarzschild solution (Schwarzschild 1916) represents the most general, spherically symmetric vacuum solution of Einstein's field equations. It describes the spacetime outside a spherical mass distribution, with no charge or angular momentum. In the so-called standard coordinates the metric coefficients read

$$A(r) = \left(1 - \frac{2m}{r}\right), \quad B(r) = 1, \quad D(r) = \frac{2m}{r}\left(1 - \frac{2m}{r}\right)^{-1}, \tag{21}$$

with $m = GM/c^2$. In this coordinate system the first-order expansion coefficients read

$$\mathcal{T}_1 = -1 \qquad \mathcal{V}_1 = -2 \qquad \mathcal{N}_1 = 3 \qquad \mathcal{H}_1 = 2 \qquad \mathcal{A}_1 = -2 \qquad \mathcal{B}_1 = 0 \qquad \mathcal{D}_1 = 2. \tag{22}$$

Note that, to calculate the PN equations of motion of order n, one always requires knowledge of $\mathcal{T}_{n+1}$. For completeness, the second-order coefficient is $\mathcal{T}_2 = 2$. The solution can also be written in other coordinate systems. We discuss several common coordinate choices and give analytical expressions for the respective expansion coefficients in Appendix A.

3. Relativistic astrometry

We now tackle the problem of measuring astrometric positions in relativistic settings. At the heart of the problem is converting a body's position vector in a particular coordinate system to a coordinate-independent quantity that any observer can agree on. This problem has been tackled over the past century using different methods (Fischbach & Freeman 1980; Epstein & Shapiro 1980; Richter & Matzner 1983; Brumberg 2017). Recently, with the advent of high-precision astrometry space missions such as Gaia (Gaia Collaboration 2016), the problem has been revisited by several authors with the aim of providing functional frameworks for high-precision astrometric measurements (Klioner & Kopeikin 1992; Kopeikin et al. 2011). Here, we show how the expansion scheme presented in Sect. 2 can be used to calculate astrometric positions in arbitrary spherically symmetric spacetimes.

3.1. Photon trajectory

Figure 1 depicts the scenario here considered. A star, $\mathcal{S}$, orbiting a massive body at the origin of our reference system emits light that is detected by an observer, O. The star and observer have coordinate vectors $\boldsymbol{r}_i$ and $\boldsymbol{r}_f$, respectively. The vector $\boldsymbol{k}$ denotes the tangent vector to the photon's trajectory at the observer's position. We also define the Euclidean distance between observer and star as $\boldsymbol{D} = (\boldsymbol{r}_f - \boldsymbol{r}_i)$ and the vector perpendicular to both $\boldsymbol{r}_i$ and $\boldsymbol{r}_f$ as $\boldsymbol{L} = \boldsymbol{r}_i \times \boldsymbol{r}_f$.

Calculating the apparent angular separation between the central object and the emitting star requires knowledge of $\boldsymbol{k}$. To evaluate it we followed the procedure presented in Brumberg (2017) and made it compatible with the above-presented general metric expansion scheme. The mathematical derivation is long, and we refer the reader to Appendix B for the detailed calculation. We outline the procedure here. We started by finding a perturbative solution for the photon's trajectory of the type

$$\boldsymbol{x}(\bar{t}) = \boldsymbol{x}_0(\bar{t}) + m\boldsymbol{x}_1(\bar{t}) + m^2\boldsymbol{x}_2(\bar{t}) + \cdots, \tag{23}$$

where $\boldsymbol{x}_0(\bar{t})$ represents the zeroth-order solution as a function of $\bar{t} = ct$, and $m = GM/c^2$ represents the gravitational radius. As we show in Appendix B, the unperturbed solution corresponds to a straight trajectory:

$$\boldsymbol{x}_0(\bar{t}) = \boldsymbol{r}_i + \boldsymbol{\sigma}\,(\bar{t} - \bar{t}_0), \tag{24}$$

where $\boldsymbol{\sigma}$ represents the tangent vector to the photon trajectory at $\bar{t} = \bar{t}_0$. We then found the general form of the first-order correction, $\boldsymbol{x}_1(t)$, and the exact trajectory connecting the initial and final points by solving the boundary-value problem. Knowing the exact form of the trajectory connecting the star and the observer immediately yields the tangent vector at the observer's position through differentiation. The result can be cast in the following form:

$$\boldsymbol{k} = (1 + \delta k_{\parallel})\,\boldsymbol{k}_{\parallel} + \delta k_{\perp}\boldsymbol{k}_{\perp}, \tag{25}$$

where $\boldsymbol{k}_{\parallel} = \boldsymbol{D}/|\boldsymbol{D}|$ represents the unit vector in the direction of the separation vector, and $\boldsymbol{k}_{\perp} = \boldsymbol{L} \times \boldsymbol{D}/|\boldsymbol{L} \times \boldsymbol{D}|$ denotes the perpendicular component to that same direction[1]. The correction terms $\delta k_{\parallel}$ and $\delta k_{\perp}$ are given by

$$\delta k_{\parallel} = -\frac{m}{|\boldsymbol{r}_f|}\left[(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1) - \frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{L}|^2}{|\boldsymbol{r}_f|^2|\boldsymbol{D}|^2}\right], \tag{26}$$

[1] Note that, because of the geometric configuration, $|\boldsymbol{L} \times \boldsymbol{D}| = |\boldsymbol{L}||\boldsymbol{D}|$.

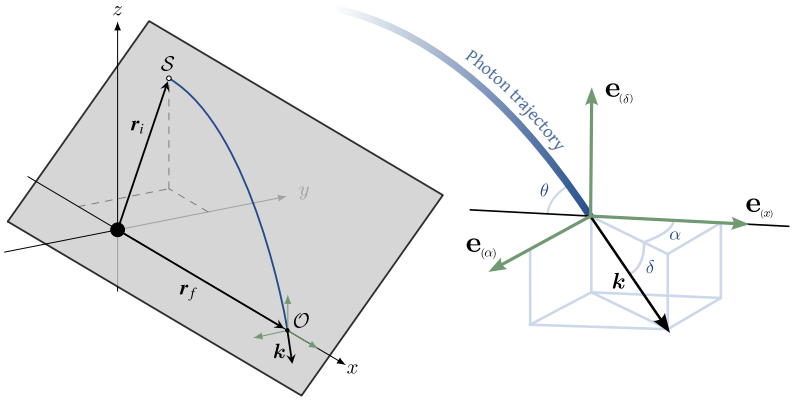


Fig. 1. Physical setting described in Sect. 3.1. *Left*: photon emitted from a star, $\mathcal{S}$, is detected by an observer, $\mathcal{O}$. The photon's trajectory (blue line) lies in the plane that contains the origin; the star's position, $\boldsymbol{r}_i$; and the observer's position, $\boldsymbol{r}_f$. $\boldsymbol{k}$ denotes the vector tangent to the photon's trajectory at the observer's location. *Right*: local tetrad at the observer's location described in Sect. 3.2 and used to calculate the photon direction as seen by the observer.

$$\delta k_\perp = -\frac{m}{|\boldsymbol{L}|}\left[\left(\mathcal{T}_1+\mathcal{V}_1+\frac{\mathcal{N}_1}{3}\right)\Big(\,|\boldsymbol{r}_f|-|\boldsymbol{r}_i|+|\boldsymbol{D}|\,\Big)+\frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{L}|^2}{|\boldsymbol{D}|^2}\left(\frac{1}{|\boldsymbol{r}_f|}-\frac{1}{|\boldsymbol{r}_i|}\right)\right]+m\frac{|\boldsymbol{D}|}{|\boldsymbol{L}|}\left[(\mathcal{T}_1+\mathcal{V}_1)\,(\mu+1)+\frac{\mathcal{N}_1}{3}(\mu^3+1)\right], \tag{27}$$

where we define $\mu = \boldsymbol{n}_f \cdot \boldsymbol{n}_D$ with $\boldsymbol{n}_f = \boldsymbol{r}_f/|\boldsymbol{r}_f|$ and $\boldsymbol{n}_D = \boldsymbol{D}/|\boldsymbol{D}|$. The above expressions give the photon-deflection direction at the observer's position for generic, spherically symmetric spacetimes. Considering the Schwarzschild metric, our formalism recovers the results of Brumberg (2017), establishing agreement with this standard reference.

3.2. Observer frame: Local tetrads and apparent angular position

To obtain the apparent position of an emitter as seen by an observer we needed to define its local inertial frame. This is done by introducing a local tetrad, $\mathbf{e}^{\mu}_{(a)}$, satisfying the condition $g_{\mu\nu}\mathbf{e}^{\mu}_{(a)}\mathbf{e}^{\nu}_{(b)} = \eta_{(a)(b)}$, where $\eta_{(a)(b)}$ represents the metric of flat spacetime. In others words, defining the locally flat frame at the observer's position. Given the metric form (2), a natural tetrad definition is

$$\mathbf{e}_{(t)} = \frac{1}{\sqrt{p(r)}}\mathbf{e}_t \qquad \mathbf{e}_{(r)} = \frac{1}{\sqrt{q(r)}}\mathbf{e}_r \qquad \mathbf{e}_{(\theta)} = \frac{1}{a(r)}\mathbf{e}_\theta \qquad \mathbf{e}_{(\phi)} = \frac{1}{a(r)\sin\theta}\mathbf{e}_\phi, \tag{28}$$

where parentheses' notation distinguishes the tetrad vectors from the coordinate basis associated with the metric. We can write the right-hand side of the equation in Cartesian form by performing the corresponding coordinate transformation. We obtain

$$\mathbf{e}_{(r)} = \frac{1}{\sqrt{q(r)}}\left[\frac{x}{r}\mathbf{e}_x+\frac{y}{r}\mathbf{e}_y+\frac{z}{r}\mathbf{e}_z\right] \qquad \mathbf{e}_{(\theta)} = \frac{1}{a(r)\sin\theta}\left[\left(\frac{zx}{r}\right)\mathbf{e}_x+\left(\frac{zy}{r}\right)\mathbf{e}_y-\left(\frac{x^2+y^2}{r}\right)\mathbf{e}_z\right] \qquad \mathbf{e}_{(\phi)} = \frac{1}{a(r)\sin\theta}\left[-y\mathbf{e}_x+x\mathbf{e}_y\right], \tag{29}$$

where we left out the time component of the tetrad as it remains unchanged. Without loss of generality we can consider the configuration in Fig. 1. Taking the observer to be at $\boldsymbol{r}_f = (r, 0, 0)$, its local tetrad reads

$$\mathbf{e}_{(x)} \equiv \mathbf{e}_{(r)} = \frac{1}{\sqrt{q(r)}}\mathbf{e}_x = \frac{1}{\sqrt{B(r)+D(r)}}\mathbf{e}_x \qquad \mathbf{e}_{(\alpha)} \equiv -\mathbf{e}_{(\phi)} = -\frac{r}{a(r)}\mathbf{e}_y = -\frac{1}{\sqrt{B(r)}}\mathbf{e}_y \qquad \mathbf{e}_{(\delta)} \equiv -\mathbf{e}_{(\theta)} = \frac{r}{a(r)}\mathbf{e}_z = \frac{1}{\sqrt{B(r)}}\mathbf{e}_z. \tag{30}$$

The measured direction of the incoming photon at the observer's position can then be calculated by projecting the components of the vector $\boldsymbol{k}$ onto the local tetrad as

$$k^{(a)} = \mathbf{e}^{(a)}_j k^j = \left(g_{ij}\,\eta^{(a)(b)}\,\mathbf{e}^i_{(b)}\right)k^j. \tag{31}$$

Using the above tetrad, the angles defined in Fig. 1 are calculated according to

$$\tan\alpha = -\frac{\mathbf{e}^{(\alpha)}_j k^j}{\mathbf{e}^{(x)}_j k^j} = \sqrt{\frac{B(r)}{D(r)+B(r)}}\,\frac{k^y}{k^x} \qquad \tan\delta = \frac{\mathbf{e}^{(\delta)}_j k^j}{\sqrt{\left(\mathbf{e}^{(x)}_j k^j\right)^2+\left(\mathbf{e}^{(\alpha)}_j k^j\right)^2}} = \frac{\sqrt{B(r)}\,k^z}{\sqrt{D(r)\,(k^x)^2+B(r)\left[(k^x)^2+(k^y)^2\right]}}, \tag{32}$$

where the minus sign in the definition of α ensures we have the correct right-ascension convention. The angle θ defined in Fig. 1 can similarly be defined as

$$\tan\theta = \frac{\sqrt{\left(\mathbf{e}^{(\alpha)}_j k^j\right)^2+\left(\mathbf{e}^{(\delta)}_j k^j\right)^2}}{\mathbf{e}^{(x)}_j k^j} = \sqrt{\frac{B(r)}{D(r)+B(r)}}\,\frac{\sqrt{(k^y)^2+(k^z)^2}}{k^x} \tag{33}$$

and represents the apparent angular separation between the lens and the star. In the far-field region, where $r \gg m$, we can use the expansion coefficients in Eq. (20) as an approximation for the metric functions. Furthermore, if the source position is close to the central object, the small angle limit holds ($k_y, k_z \ll k_x$) and the expressions simplify to their Euclidean limit:

$$\alpha = \frac{k^y}{k^x}, \qquad \delta = \frac{k^z}{k^x}, \qquad \theta = \frac{\sqrt{(k^y)^2 + (k^z)^2}}{k^x}. \tag{34}$$

Note that in this limit $\theta^2 = \delta^2 + \alpha^2$.

The above equations represent the apparent position of the star for a static observer. In the most general scenario, the observer will not be static. The Solar System, for example, revolves around the centre of the galaxy with a velocity of ~230 km/s corresponding to $\beta = v/c \sim 10^{-3}$. In this case, we need to calculate the observed direction of the photon as seen by a moving observer. This is achieved through a Lorentz transformation. If the observer is moving with velocity, $\boldsymbol{\beta}$, the direction of the photon observed by the moving observer, $\boldsymbol{k}'$, is given by (Kopeikin et al. 2011)

$$\boldsymbol{k}' = \frac{1}{\gamma\,(1 - \boldsymbol{k}\cdot\boldsymbol{\beta})}\left[\boldsymbol{k} + \boldsymbol{\beta}\left(\frac{(\gamma - 1)\,(\boldsymbol{\beta}\cdot\boldsymbol{k})}{\beta^2} - \gamma\right)\right], \tag{35}$$

with $\gamma = (1 - \beta^2)^{-1/2}$ representing the Lorentz factor. This expression can thus be applied to $\boldsymbol{k}$ before plugging its components into the formulas in Eq. (32). This yields the observed apparent position of the star as seen by the moving observer.

4. Relativistic observables

Observations of stars in the GC deliver time-resolved astrometric and spectroscopic measurements. Having established the formalism for modelling the astrometric position of an emitter in the previous section, we now turn to the remaining observables required for a complete relativistic description of these observations: the arrival time and the frequency shift between emission and detection.

4.1. Time delay

The arrival time of photons carries information about the total time delay accumulated along the photon path. Knowing the exact path taken by the photon is thus key for this calculation. From the perturbative form of the photon's trajectory in Eq. (23), we note that the norm of the separation vector between the emitter and observer, $\boldsymbol{D}$, can be written as

$$|\boldsymbol{D}| = (\bar{t}_f - \bar{t}_0) + m(\boldsymbol{x}_1 \cdot \boldsymbol{\sigma}) + O(m^2). \tag{36}$$

Using the exact expression for $\boldsymbol{x}_1$ derived in Appendix B (see Eqs. (B.19) and (B.18)), we can solve for the arrival time $\bar{t}$, obtaining

$$\bar{t} = \bar{t}_0 + |\boldsymbol{D}| - m\left[\mathcal{A}(\boldsymbol{r}_f) - \mathcal{A}(\boldsymbol{r}_i)\right], \tag{37}$$

where

$$\mathcal{A}(\boldsymbol{r}) = \frac{\mathcal{N}_1}{3}\frac{(\boldsymbol{r}\cdot\boldsymbol{\sigma})}{|\boldsymbol{r}|} - (\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1)\log(\boldsymbol{r}\cdot\boldsymbol{\sigma} + |\boldsymbol{r}|). \tag{38}$$

When we consider the coefficients for the Schwarzschild metric in isotropic coordinates, the above expression reduces to the classical Shapiro delay (Shapiro 1964; Brumberg 2017).

4.2. Redshift

The second measurable quantity from the detected light is the redshift of specific spectral features. Understanding how these features change between their emission and their detection requires connecting the emitter and observer frames. A monochromatic signal can be described as a wave with phase φ, and the frequency measured by any observer is its rate of change with respect to proper time. Denoting the proper time by ℓ, we write $\nu = (d\varphi/d\ell)$. Because φ is a scalar, it will be a constant along the whole trajectory, and the ratio of the frequency emitted and observed will be the ratio of the emitter and observer proper times. This implies

$$\frac{\nu_0}{\nu} = \frac{d\ell}{d\ell_0} = \frac{(d\ell/d\bar{t})}{(d\ell_0/d\bar{t}_0)}\left(\frac{d\bar{t}}{d\bar{t}_0}\right) = \frac{(ds/d\bar{t})_{\boldsymbol{r}(\bar{t})}}{(ds/d\bar{t})_{\boldsymbol{r}(\bar{t}_0)}}\left(\frac{d\bar{t}}{d\bar{t}_0}\right), \tag{39}$$

where ν_0 and ν denote the emitted and observed frequency, respectively. The terms in the fraction can be obtained directly from the metric form in Eq. (3). The numerator and denominator read

$$\left(\frac{ds}{d\bar{t}}\right) = \sqrt{-A(r) + \beta^2 B(r) + D(r)\,(\boldsymbol{n}\cdot\boldsymbol{\beta})^2}, \tag{40}$$

which is evaluated at the respective points of emission and detection using the exact expressions for $A(r)$, $B(r)$, and $D(r)$. To evaluate the second term, we used the time delay calculated in Eq. (37) to write

$$\frac{d\bar{t}}{d\bar{t}_0} = 1 + \frac{d}{d\bar{t}_0}\Big(\,|\boldsymbol{D}| - m[\mathcal{A}(\boldsymbol{r}) - \mathcal{A}(\boldsymbol{r}_i)]\,\Big) = 1 + \frac{d\bar{t}}{d\bar{t}_0}\frac{d}{d\bar{t}}\Big(\,|\boldsymbol{D}| - m[\mathcal{A}(\boldsymbol{r}) - \mathcal{A}(\boldsymbol{r}_i)]\,\Big) \tag{41}$$

and obtain

$$\frac{d\bar{t}}{d\bar{t}_0} = \left[1 - \frac{d}{d\bar{t}}\Big(|\boldsymbol{D}| - m[\mathcal{A}(\boldsymbol{r}) - \mathcal{A}(\boldsymbol{r}_i)] \Big)\right]^{-1} = \left[1 - \frac{\boldsymbol{D}}{|\boldsymbol{D}|}\cdot\frac{\big(\boldsymbol{v}_f - \boldsymbol{v}_o\big)}{c} + m[\dot{\mathcal{A}}(\boldsymbol{r}) - \dot{\mathcal{A}}(\boldsymbol{r}_i)]\right]^{-1}. \tag{42}$$

The notation $\dot{\mathcal{A}}$ denotes a derivative with respect to τ, and using Eq. (38) we have

$$\dot{\mathcal{A}}(\boldsymbol{r}) = \frac{\boldsymbol{v}\cdot(\boldsymbol{\sigma}+\boldsymbol{n})}{|\boldsymbol{r}|}\cdot\left[\frac{\mathcal{N}_1}{3} - \frac{(\mathcal{T}_1+\mathcal{V}_1+\mathcal{N}_1+\mathcal{H}_1)}{1+\boldsymbol{n}\cdot\boldsymbol{\sigma}}\right] - \frac{\boldsymbol{v}\cdot\boldsymbol{n}}{|\boldsymbol{r}|}\cdot\left[\frac{\mathcal{N}_1}{3}\Big(1+\boldsymbol{n}\cdot\boldsymbol{\sigma}\Big)\right], \tag{43}$$

where we used the notation $\boldsymbol{n} = \boldsymbol{r}/|\boldsymbol{r}|$. Note that this expression depends on the initial direction of propagation, $\boldsymbol{\sigma}$. However, to first order, $\boldsymbol{\sigma} = \boldsymbol{D}/|\boldsymbol{D}|$. Furthermore, when the observer distance is such that $|\boldsymbol{r}_f| \gg m$, we can reduce the calculation to that of $\dot{\mathcal{A}}(\boldsymbol{r}_i)$. This yields a concise expression to evaluate Eq. (39). Note as well that, to zeroth order, the ratio of proper times is

$$\frac{d\tau}{d\tau_0} = 1 + \frac{\boldsymbol{D}}{|\boldsymbol{D}|}\cdot\frac{\big(\boldsymbol{v}_f - \boldsymbol{v}_o\big)}{c} + O(m), \tag{44}$$

which is classical redshift caused by the differential velocity along the line of sight.

5. Application to the Galactic centre

The centre of the Milky Way represents the best laboratory to study gravity near supermassive compact objects (Genzel et al. 2010). The framework presented above is particularly well suited to model the observation of stars moving deep within the gravitational potential of Sgr A*. We showcase its applicability to the best dynamical probe of spacetime around Sgr A*, the star S2 (GRAVITY Collaboration 2018, 2020).

The star S2 follows a highly eccentric orbit around Sgr A* with a period of approximately 16 years. It has a semi-major axis of $\sim 10^3$AU and an eccentricity of $e = 0.88$. The closest point to Sgr A* along its orbit is at only 120AU. For the central object's mass of $4.2\times10^6 M_\odot$, this corresponds to $\sim 2.8\times10^3$ gravitational radii, placing it firmly in the regime where PN corrections to the photon trajectory can play an important role. In this section we denote the gravitational radius by $R_g = GM/c^2$.

5.1. Gravitational lensing

We can assess the magnitude of the gravitational-lensing correction to the apparent position of S2 as predicted by the framework derived in the previous sections. We used the orbital parameters presented in GRAVITY Collaboration (2020) to calculate the initial position and velocity of the star at apocentre and integrated the orbit using the 1PN equations of motion derived in Sect. 2.1. We then calculated the observed position using the method derived in Sect. 3. We worked with the Schwarzschild metric in standard coordinates and took a distance to the GC of $R_0 \sim 8.27$ kpc.

Figure 2 compares the integrated orbit of S2 when the PN formalism of Sect. 3 is (or is not) considered for the calculation of the apparent position of the star as seen from Earth. In particular, after integrating the orbit we calculated the apparent position of the star using the definitions in Eq. (34). When the light deflection is not considered, the apparent position is given by the parallax limit formulas. If the star has coordinates (x, y, z) and the observer is placed at coordinates $(R_0, 0, 0)$, the apparent angular separation from Sgr A* is

$$\alpha_{\rm parallax} = \frac{y}{R_0 - x}, \qquad \delta_{\rm parallax} = \frac{z}{R_0 - x}, \qquad \theta_{\rm parallax} = \frac{\sqrt{y^2+z^2}}{R_0 - x} \equiv \frac{\rho}{R_0 - x}, \tag{45}$$

where we used the axis convention of Fig. 1 and defined the impact parameter $\rho = \sqrt{y^2+z^2}$. The lensing causes the orbit to appear larger by a few tens of μas. The maximum deviation occurs close to the pericentre passage with a difference between the lensed and non-lensed position of $\sim$30 μas.

We can also compare our results with the thin-lens approximation typically used in cosmological or microlensing studies (Schneider et al. 1992). The thin-lens approximation assumes that all the light deflection happens in the plane of the lens. In our case, this would be the plane $x = 0$. The apparent angular separation between the star and Sgr A* in this approximation is given by

$$\alpha_{\rm thin\text{-}lens} = \frac{y}{\sqrt{y^2+z^2}}\,\theta_{\rm thin\text{-}lens}, \qquad \delta_{\rm thin\text{-}lens} = \frac{z}{\sqrt{y^2+z^2}}\,\theta_{\rm thin\text{-}lens}, \qquad \theta_{\rm thin\text{-}lens} = \frac{1}{2}\left[\,\theta_{\rm parallax} + \sqrt{\theta^2_{\rm parallax} + 4\theta^2_{\rm Einstein}}\,\right], \tag{46}$$

where the Einstein radius, $\theta_{\rm Einstein}$, is defined as

$$\theta_{\rm Einstein} = \sqrt{\frac{4R_g}{R_0}}\sqrt{\frac{-x}{R_0 - x}}. \tag{47}$$

Note how the expression is only valid when the star is behind the lens plane ($x < 0$). When in front of the plane, we used the parallax angle instead. To evaluate the quality of each approximation we numerically calculated the deflection angle as predicted by the exact equations of motion for photons derived in Sect. 2.2. In particular, for an observer at coordinates $(R_0, 0, 0)$, we shot a photon at an

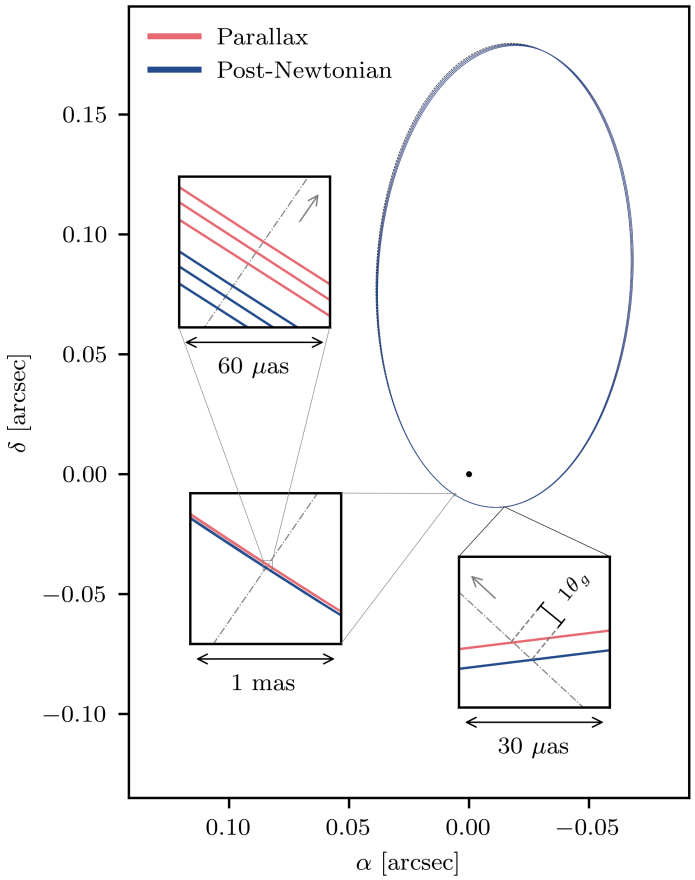


Fig. 2. Apparent position of the star S2 with and without taking into account light bending effects (blue and red lines, respectively). The orbit is integrated using the equations of motion in standard coordinates and the light deflection using the far-field approximation in Eq. (34). The initial position and velocity are taken to be the one at the apocentre of the orbit characterised by the orbital elements in GRAVITY Collaboration (2020). The insets on the left zooming in on the pericentre passage. The inset in the lower right zooms in on the point where the star crosses the plane perpendicular to the direction of Sgr A* (y–z plane of Fig. 1). At this point, the lensed position is shifted by $\theta_g = R_g/R_0$ from the non-lensed position. The grey arrows point towards Sgr A*.

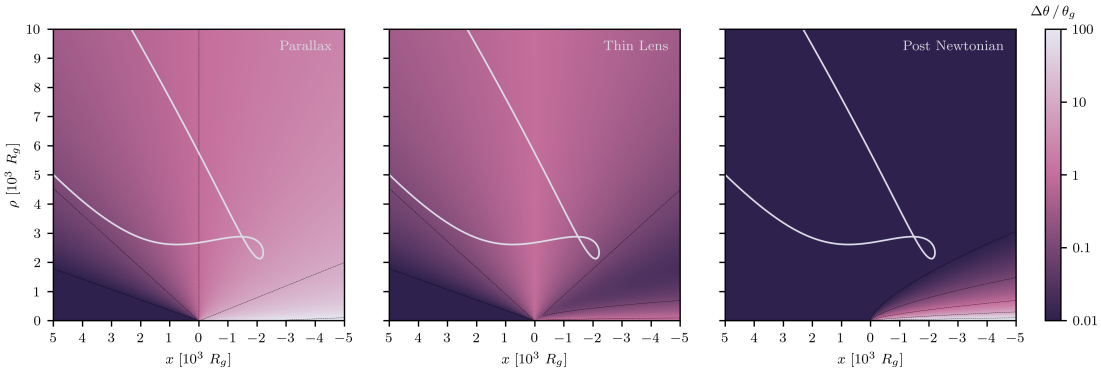


Fig. 3. Comparison between analytical approximations of the deflection angle and the numerically ray-traced result. *From left to right*: error incurred by using the parallax, thin-lens, and PN approximations in units of $\theta_g = R_g/R_0$. The observer is placed along the x-axis at a distance of $R_0 = 10^8\ R_g$ and the mass of Sgr A* is taken to be $M = 4.2 \times 10^6\ M_\odot$. We performed the numerical experiment for a wide range of observers distances. The actual distance to Sgr A* is $\sim 10^{10} R_g$, but at distances larger than $10^8 R_g$ no quantitative or qualitative differences are seen.

angle of θ_0 with respect to the radial direction. This defines a unique geodesic for which all points are observed at an angle of θ_0. For each coordinate point along the geodesic, we then calculated the corresponding angle in each approximation: parallax, thin-lens, and PN. The result is shown in Fig. 3. The plots show the deviation from the numerical ray-tracing method in units of $\theta_g = R_g/R_0$, corresponding to the angular size of the gravitational radius. For Sgr A*, we have $\theta_g \sim 5$ µas. The white line corresponds to the orbit of S2.

The first striking feature is the accuracy of the PN method when compared to the other approximations. Note, however, that the approximation breaks down in a parabolic-shaped region behind Sgr A*. The PN approximation implicitly assumes the zeroth-order solution is a straight line. The more closely the photons graze the central object, the more we deviate from this assumption.

Despite a crude simplification, the thin-lens approximation shows a relatively high level of agreement with the numerical results, with errors below θ_g. The approximation shows the highest disagreement with the numerical solution in the $x = 0$ plane where the assumption of a large source distance from the lens plane breaks down. A conical region behind Sgr A* is particularly well described by this approximation where the assumptions of the thin-lens equation hold.

A final note on the current accuracy of the GRAVITY instrument is warranted. Currently, the typical astrometric error on the measured positions of S2 are of ∼30 µas (GRAVITY Collaboration 2020). In this regime, the parallax approximation is still a

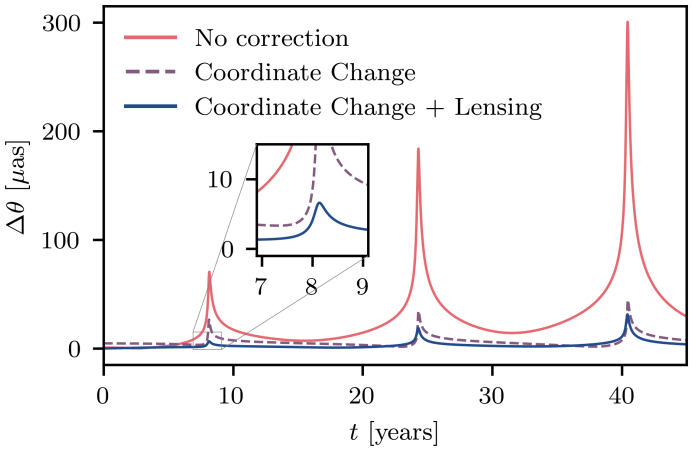


Fig. 4. Absolute difference in astrometric position between the integrated orbit in harmonic and standard coordinates when the initial radial coordinate is interpreted in the same way in the two systems and no light bending is considered (red line), when the appropriate coordinate transformation is applied (dashed purple line), and when both this and light bending are taken into account (solid blue line). When the proper coordinate transformation and light bending are considered the difference between coordinate systems stays below the approximation order.

suitable approach to modelling the astrometric data and reassures the analysis performed so far by the GRAVITY Collaboration. The expected (and already real) improvement in sensitivity with the GRAVITY+ upgrade (GRAVITY+ Collaboration 2022) pushes the astrometric limits to the point where either the thin-lens or, preferably, the PN approximation must be adopted for a reliable modelling of the astrometric data.

5.2. Coordinate ambiguities

Observational studies of the star S2 have, so far, ignored the light deflection effects just discussed and rely on the parallax approximation to project the physical position of the star in spacetime to the apparent angular position as seen by an observer. The problem with this approach goes beyond the incurred error studied in the previous section. The problem lies in the classical question of what is an observable in GR. Unlike in a Newtonian view of the Universe, where physical coordinates such as the position of a star are absolute and measurable quantities, no such absolute reference exists in GR. We must compare our observations with coordinate independent quantities such as proper time intervals, frequency ratios, or angular separations[2]. Here we show the impact of neglecting such subtle aspects from astrometric analysis and how these can lead to discrepancies between studies.

The main analysis is summarised in Fig. 4. We defined the initial apocentre position and velocity using the orbital parameters of S2 and integrated the orbit using both standard and harmonic coordinates. The difference between the angular separation of S2 and Sgr A* calculated from both coordinate systems when the parallax approximation is used is represented by the red line. The disagreement can reach up to 60 μas per orbital revolution, with the total difference between the two coordinate systems reaching 0.3 mas after three orbital revolutions. The reason for the discrepancy stems from implicitly assuming that the radial coordinate in both coordinate systems can be interpreted in the same way. Although they both represent a radial coordinate, they are related by a coordinate transformation (see Appendix A). When the initial apocentre position is transformed into the harmonic coordinate system, a better agreement is found (dashed purple line), and when the PN approximation is further considered the difference can be made to fall below the approximation order. As the precision in our measurements increases, so must our modelling. Comparing distinct studies where no light bending is considered requires an understanding of the underlying coordinate system used for the analysis. The PN approximation and the associated tetrad formalism solves both of these problems by enforcing a coordinate choice and calculating measurable quantities.

5.3. Testing metric theories

As mentioned in Sect. 2, spherically symmetric models of spacetime provide a natural setting in which to study a wide class of gravity theories. Amongst the large family of alternative theories to GR, many admit spherically symmetric solutions for which the metric parameters $p(r)$, $q(r)$, and $a(r)$ in Eq. (2) depend on theory-dependent parameters. The parametrised post-Newtonian (PPN) framework grew out of a long tradition of efforts to place empirical constraints on metric theories of gravity without requiring a specific theory to be named. With its conceptual roots tracing back to the pioneering work of Eddington (Eddington 1923) and later Robertson and Schiff (Robertson 1962; Schiff 1967), the framework was placed in its systematic and contemporary form by Nordtvedt and Will (Nordtvedt 1970; Will 1971), who introduced a comprehensive set of parameters capable of capturing deviations from general relativity across a large family of competing theories. For the class of spherically symmetric spacetimes, a connection with the expansion method of the current manuscript can be made. Following a similar approach to Sarmiento (1982) we write the metric coefficients as

$$p(r) = \left(1 - \alpha\frac{2m}{r} + \beta\frac{2m^2}{r^2} - \eta\frac{2m^3}{r^3}\right), \qquad q(r) = \left(1 + \gamma\frac{2m}{r} + \delta\frac{4m^2}{r^2} + \lambda\frac{8m^3}{r^3}\right), \qquad a^2(r) = r^2 q(r). \tag{48}$$

Note that our choice of $a(r)$ renders the spatial part of the metric conformally flat. This is in contrast with Sarmiento (1982), but it allows a direct comparison of the derived equations of motion with the more common isotropic form presented in the literature (Will 1993; Poisson & Will 2014).

[2] We refer the reader to chapter 6 of Brumberg (2017) for a comprehensive discussion of the topic.

The two classical PPN parameters are γ and β, each admitting a clear physical interpretation: γ quantifies the degree of spatial curvature induced by a unit rest mass, while β characterises the non-linearity of the gravitational superposition law. Note that α can be absorbed by the scale parameter m, and thus be set to one. The first order, the expansion coefficients read

$$\mathcal{T}_1 = -1 \qquad \mathcal{V}_1 = -\gamma \qquad \mathcal{N}_1 = 0 \qquad \mathcal{H}_1 = 2(1+\gamma) \qquad \mathcal{A}_1 = -2 \qquad \mathcal{B}_1 = 2\gamma \qquad \mathcal{D}_1 = 0. \tag{49}$$

For completeness, we must also state the second-order coefficient $\mathcal{T}_2 = 2(\beta + \gamma)$. These match the usual PN prescription seen in the literature. Note that, to first order, the β parameter uniquely affects the orbital dynamics of a star and not the motion of a travelling photon, as also seen on the PPN formulas for the light deflection and the Shapiro time delay (Will 2014). We present higher order coefficients in Appendix C.

The exact values of the PPN parameters are theory dependent. For example, the well-known Brans–Dicke theory (Brans & Dicke 1961) whose Lagrangian depends a the coupling parameter ω_{BD} is prescribed by (Nordtvedt 1970)

$$\beta = 1, \qquad \gamma = \frac{1+\omega_{BD}}{2+\omega_{BD}}. \tag{50}$$

The prescription of more general scalar-tensor theories where the coupling constant becomes a function of a dynamical scalar field as $\omega(\phi)$, require a more elaborate prescription but the PPN parameters can be written as (Will 1993)

$$\beta = 1 + \frac{d\omega/d\phi}{(3+2\omega)(4+2\omega)^2}, \qquad \gamma = \frac{1+\omega}{2+\omega}. \tag{51}$$

The above examples show the β and γ parameters, but the corresponding value of the higher order PPN parameters δ and λ have so far not been derived in the literature. Other alternatives to GR-like $f(R)$-gravity or the more general Horndeski theory can similarly be mapped but require the exact matching of their PN expansion with the above-derived PN metric. This can be a tedious and challenging process. Nonetheless, by virtue of the PPN metric, the metric parameters can be agnostically compared to data and mapped to the corresponding theories post facto.

5.4. Testing the black-hole paradigm

Besides testing metric theories of gravity, the expansion formalism also allows us to test specific metric solutions of Einstein's equations besides the well-established Schwarzschild solution. We highlight here its application to the case of a charged BH described by the Reissner–Nordström metric (Reissner 1916; Nordström 1918) due to its simplicity and show how the charge of Sgr A* can, in principle, be constrained with the accuracy of current GRAVITY observations.

The Reissner–Nordström metric is the simplest physically based extension of the Schwarzschild metric and can be written as

$$p(r) = \left(1 - \frac{2m}{r} + \frac{2q^2}{r^2}\right), \qquad q(r) = \frac{1}{p(r)}, \qquad a(r) = r, \tag{52}$$

where the charge parameter, q, is defined as

$$q^2 = \frac{Q^2 G}{4\pi\epsilon_0 c^4}, \tag{53}$$

with Q representing the charge of the body in Coulomb and ϵ_0 the vacuum permittivity. One can naturally expand the above expressions within the PN framework by defining the dimensionless charge parameter $\Theta^2 = q^2/m^2$. Appendix D shows how, for this particular form of the metric, analytical expressions for the expansion coefficients can be obtained. We took a different approach in this section to make the connection with the PPN metric clear. We first transformed the metric to isotropic coordinates and only then performed the expansion. To make our discussion clearer we compile the calculations in Appendix E and highlight the result. When expanded to second order, the isotropic form of the RN metric reads

$$p(r) = 1 - \frac{2m}{r} + \left(1+\Theta^2\right)\frac{2m^2}{r^2} - \left(\frac{3-10\Theta^2}{4}\right)\frac{2m^3}{r^2}, \qquad q(r) = 1 + \frac{2m}{r} + \left(\frac{3-2\Theta^2}{8}\right)\frac{4m^2}{r^2}, \qquad a^2(r) = r^2 q(r), \tag{54}$$

from which we can immediately extract the equivalent PPN parameters of the metric (48). For the first-order parameters, β and γ, we have

$$\beta = 1 + \Theta^2 \quad \gamma = 1. \tag{55}$$

This highlights how, by using the PPN formalism, we can not only constrain alternative metric theories, but also other metric solutions. Distinct solutions will, analogously to alternative gravity theories, be described by a particular set of PPN parameters.

For the case of a charged BH, we can quantify the observational effects on the orbit of stars. Figure 5 displays integrated orbit of the star S2 for several values of the charge parameter. The net effect on the orbit is to reduce the observed precession as the charge is increased. For a charge parameter of $\Theta = 0.6$, the observed difference from a non-charged BH can be as large as 100 μas, which is well within the current precision of GRAVITY observations.

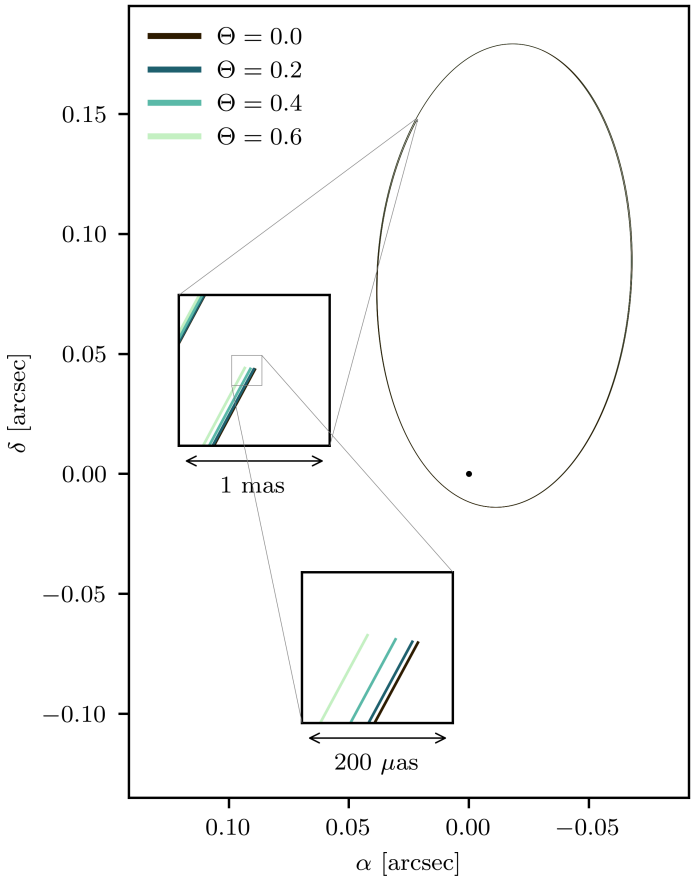


Fig. 5. Apparent position of the star S2 if orbiting a charged BH for several values of the charge parameter, Θ, defined in Sect. 5.4. The orbit is integrated similarly to what was done in Fig. 2. The insets zoom in on the region where the effects of the different orbital precession rates are more pronounced. The black circle in the lower inset illustrates the scale of the problem and represents the projected size of the event horizon of Sgr A*.

6. Discussion

Our results highlight the importance of self-consistently modelling the observational data of stars orbiting Sgr A* and how the PN formalism is well suited to this end. In its traditional form, the PN formalism is concerned with the generic problem of describing both the motion of extended bodies in spacetime and the dynamics of spacetime itself (Poisson & Will 2014). It has a long and successful history of application across a wide range of fields, from satellite communications to the most extreme astrophysical settings. In the Solar System, the mathematical foundations for high-precision astrometry, navigation, and timekeeping rely heavily on the PN formalism (Kopeikin et al. 2011), and its importance in the broader context of astrophysics was established by the International Astronomical Union in Soffel et al. (2003). The outlined methodology in those documents is of extreme importance for Very Large Baseline Interferometry (Petit & Luzum 2010), Pulsar timing experiments (Taylor et al. 1979) and high-precision astrometric missions such as Gaia (Gaia Collaboration 2016), where a full relativistic treatment of observational quantities is required.

The first direct detection of a gravitational wave (Abbott et al. 2016) has also highlighted the importance of the PN formalism in the study of compact binaries and their gravitational-wave emission. In these settings, where conservative dynamics and back-reaction effects must be consistently accounted for, the PN formalism enables a systematic study of these effects. Despite the difficulty in performing such calculations at high PN orders, the equations of motion for non-spinning binaries are currently known up to the 4PN order (Bernard et al. 2018). The importance of this programme cannot be understated. Current and future gravitational detectors rely critically on theoretical waveform templates obtained through the PN formalism (Blanchet 2024).

In the GC, the importance PN effects in the analysis of radial velocity data became evident in the early 2000s with the improved spectroscopic capabilities delivered by near infrared field spectrometers (Eisenhauer et al. 2003; Larkin et al. 2006). These are routinely taken into account and their effects are well established (Zucker et al. 2006).

In this work, we explored the impact and importance of incorporating the PN formalism in the analysis of high-precision astrometric data of stars in the GC. The outlined results show how light bending and coordinate choices play a subtle but crucial role when modelling such observations, and that, although these effects are below the current statistical error of observational data (GRAVITY Collaboration 2020), they become increasingly important with new-generation instruments and the long-term goal of accurately tracking the position of the S-stars (GRAVITY+ Collaboration 2022).

Our results build on the large body of work on the PN formalism. Section 2 highlights one key methodological difference between our work and other approaches in the literature. Rather than performing the PN expansion at the metric level, we applied it directly to the equations of motion and keep their functional form explicit. Although mathematically equivalent, our approach highlights the functional form of the PN equations of motion, which are of central importance for the study of stars orbiting Sgr A*. Sections 3 and 4 then derive observable quantities within this expansion scheme, and Sect. 5 demonstrates its applicability to the GC. In particular, we show how this expansion scheme connects to the PPN formalism and how it can be used to test deviations from GR through both the PPN formalism and specific metric solutions to Einstein's field equations.

It is important to note that we restricted our analysis to spherically symmetric solutions and the corresponding leading-order PN corrections to both the orbital motion of the S-stars and their apparent positions as measured by an Earth-based observer. This reflects the current observational limits of GC observations. Higher order PN corrections remain beyond the reach of current observational efforts. Rigorously testing the BH paradigm will ultimately, however, require a self-consistent treatment of higher order effects. In particular, axisymmetric solutions such as the Kerr solution, introduce corrections to the equations of motion at higher PN order (Misner et al. 2008; Wex & Kopeikin 1999), and a complete analysis of S-star data must self-consistently take these into account. Most importantly, measuring the Lens–Thirring precession and the quadrupole moment of Sgr A* requires 1.5PN and 2PN models (Will 2008), respectively. Achieving this level of modelling precision will be essential as observational capabilities continue to improve, and our study represents a step in that direction.

Despite the focus on the GC, our results can be connected to more general results within the PN formalism. The equations of motion derived in Eq. (11) for massive particles and in Eq. (16) for massless particles represent exact expressions for the two-body relativistic problem when one of the masses is infinitely small. Knowing the exact form of the equations of motion in this test-body limit provides stringent benchmarks for the two-body scenario. Consequently, deriving the precise PN dynamics in the test-body limit, as presented here, supports the broader programme of developing high-accuracy binary models relevant to gravitational-wave observations.

7. Conclusion

Despite more than a century having passed since the publication of Einstein's field equations (Einstein 1915), the theoretical and observational efforts to find its limits continue (Will 2014). In this context, the PN formalism has had a long and successful history. Both within and outside the Solar System, it has been used to place the tightest constraints on deviations from GR across a wide range of gravitational systems (Fomalont et al. 2009; Bruns 2018; Shapiro et al. 2004; Lopresto et al. 1991; Taylor et al. 1979; Kramer et al. 2021).

The status of Sgr A* as the closest supermassive compact object to Earth makes the GC a uniquely privileged laboratory, probing gravitational field strengths largely inaccessible to Solar System tests or pulsar timing. Yet, only the recent advances in optical interferometry have brought the astrometric precision of GC observations to a level comparable to these other systems. Even so, effects such as light bending have thus far played a minor role in interpreting the data, owing to their subdominant magnitude at current precisions. With the advent of next-generation instruments such as GRAVITY+ (GRAVITY+ Collaboration 2022) and the Extremely Large Telescope (Padovani & Cirasuolo 2023), a self-consistent relativistic analysis of observations is essential. The formalism outlined in this manuscript provides precisely this framework, laying down the machinery to pursue these tests in the era of high-precision near-infrared interferometry.

Acknowledgements. The authors thank the anonymous referee for a careful reading of the manuscript and for remarks that contributed to improving its final form. D.C. Ribeiro thanks Laura Bernard and Re'em Sari for fruitful discussions and feedback on the manuscript. The GRAVITY+ Collaboration is a close partnership between the Max Plank for Extraterrestrial Physics (MPE), the Laboratory of Space Studies and Instrumentation in Astrophysics (LIRA) of the Paris Observatory/CNRS/Sorbonne Université/Université Paris Cité, the Institut de Planétologie et d'Astrophysique de Grenoble (IPAG) of the Université Grenoble Alpes/CNRS, the Max Planck Institute for Astronomy (MPIA), the University of Cologne, the Centro de Astrofisica e Gravitação (CENTRA), the University of South Hampton, the Institute National Des Sciences De L'Univers (INSU), the Katholieke Universiteit Leuven (KU Leuven) and the European Southern Observatory (ESO). We are very grateful to our funding agencies (MPG, ERC, CNRS [PNCG, PNGRAM], DFG, BMBF, Paris Observatory [CS, PhyFOG], Observatoire des Sciences de l'Univers de Grenoble, and the Fundação para a Ciência e Tecnologia), to ESO and the Paranal staff, and to the many scientific and technical staff members in our institutions. The research leading to these results has received funding from the European Research Council (ERC) under the European Union's Horizon 2020 research and innovation program (project UniverScale, grant agreement 951549). J.S-B. acknowledges the support received by the UNAM DGAPA-PAPIIT project IG-101025 and from the SECIHTI Ciencia de Frontera project CBF-2025-I-3033. V.C., C.C., P.G. and N.M thank the Fundação para a Ciência e Tecnologia (FCT) for the financial support to the Center for Astrophysics and Gravitation (CENTRA/IST/ULisboa) through grant No. UID/PRR/00099/2025 (https://doi.org/10.54499/UID/PRR/00099/2025) and grant No. UID/00099/2025 (https://doi.org/10.54499/UID/00099/2025).

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Appendix A: The Schwarzschild space-time

The Schwarzschild metric represent the unique vacuum solution to Einstein's field equations that is spherically symmetric and asymptotically flat (Jebsen 1921; Birkhoff & Langer 1923). In its standard from, the Cartesian metric coefficients associated with Eq. (3) read

$$A(r) = \left(1 - \frac{2m}{r}\right), \quad B(r) = 1, \quad D(r) = \frac{2m}{r}\left(1 - \frac{2m}{r}\right)^{-1}. \tag{A.1}$$

Other coordinate systems can be considered giving rise to different functional forms of these functions. Two particular commonly used choices are the isotropic and harmonic coordinates. These are related to the standard coordinate system by a direct change of the radial coordinate. The isotropic radial coordinate, ρ, is related to the standard radial coordinate, r, by means of the transformation

$$r = \rho\left(1 + \frac{m}{2\rho}\right)^2. \tag{A.2}$$

The corresponding metric functions are

$$A(\rho) = \left(\frac{1 - m/2\rho}{1 + m/2\rho}\right)^2, \quad B(\rho) = \left(1 + \frac{m}{2\rho}\right)^4, \quad D(\rho) = 0. \tag{A.3}$$

Note that the naming the radial coordinate ρ is convention. We could just as easily use the letter r to refer to the radial coordinate. Similarly, the harmonic radial coordinate, $\bar{r}$, is related to the standard radial coordinate according to

$$\bar{r} = r - m, \tag{A.4}$$

for which the metric functions read

$$A(\bar{r}) = \left(\frac{1 - m/\bar{r}}{1 + m/\bar{r}}\right), \quad B(\bar{r}) = \left(1 + \frac{m}{\bar{r}}\right)^2, \quad D(\bar{r}) = \left(\frac{m}{\bar{r}}\right)^2\left(\frac{1 + m/\bar{r}}{1 - m/\bar{r}}\right). \tag{A.5}$$

The harmonic coordinate system is particularly important within the PN formalism since it allows several symmetries to be exploited (Poisson & Will 2014). The expansion coefficients of the metric functions in these three coordinate systems can be written down analytically and thus, can be easily evaluated at any PN order. Table A.1 displays the expansion coefficients in Eqs. (11), (16), and (19) for the three coordinate systems. As mentioned in Sect. 6, these coefficients can be directly compared with the two-body PN problem when one of the masses is set to zero. In fact, the 2PN coefficients in harmonic coordinates match those found in Mirshekari & Will (2013) and Bernard (2019) when the secondary mass is zero.

Appendix B: Boundary value problem in spherically symmetric space-times

The light deflection formula presented in Eqs. (26) and (27) was obtained in two steps. First we found the general equation describing the trajectory of a photon given an initial position and emission direction. Then, we solved the boundary value problem connecting the emission point with the observation one.

B.1. Photon's trajectory

To find the generic equation describing the trajectory of the photon we considered the expansion form of the solution in powers of the gravitational radius, $m = GM/c^2$,

$$\boldsymbol{x}(\bar{t}) = \boldsymbol{x}_0(\bar{t}) + m\boldsymbol{x}_1(\bar{t}) + O(m^2), \tag{B.1}$$

and truncated it at first order since we restricted our analysis to 1PN corrections. By plugging this expansion form in the PN Eqs. (16) and (19) and gathering same order terms we can explicitly write the equations of motion as

$$\left(\frac{d^2\boldsymbol{x}_0}{d\bar{t}^2}\right) + m\left(\frac{d^2\boldsymbol{x}_1}{d\bar{t}^2}\right) = \left(\frac{d^2\boldsymbol{x}}{d\bar{t}^2}\right)_{0\mathrm{PN}} + \left(\frac{d^2\boldsymbol{x}}{d\bar{t}^2}\right)_{1\mathrm{PN}}. \tag{B.2}$$

By virtue of the asymptotic flatness condition of the expansion coefficients (12), the 0PN acceleration is null and the zeroth order solution must satisfy

$$\frac{d^2\boldsymbol{x}_0}{d\bar{t}^2} = 0, \tag{B.3}$$

whose solution is, as expected, a straight line

$$\boldsymbol{x}_0(\bar{t}) = \boldsymbol{r}_i + \boldsymbol{\sigma}(\bar{t} - \bar{t}_0), \tag{B.4}$$

Table A.1. Expansion coefficients of the Schwarzschild metric in standard, isotropic, and harmonic coordinates.

	n	1	2	3	4	5	6	7	8	⋯	k
Standard	$\mathcal{T}_n$	-1	2	0	0	0	0	0	0	⋯	0
	$\mathcal{V}_n$	-2	0	0	0	0	0	0	0	⋯	0
	$\mathcal{N}_n$	3	2	4	8	16	32	64	128	⋯	2^{k-1}
	$\mathcal{H}_n$	2	4	8	16	32	64	128	256	⋯	2^k
	$\mathcal{A}_n$	-2	0	0	0	0	0	0	0	⋯	0
	$\mathcal{B}_n$	0	0	0	0	0	0	0	0	⋯	0
	$\mathcal{D}_n$	2	4	8	16	32	64	128	256	⋯	2^k
Isotropic	$\mathcal{T}_n$	-1	4	$-\frac{35}{4}$	14	$-\frac{147}{8}$	21	$-\frac{693}{32}$	$\frac{165}{8}$	⋯	$\left(-\frac{1}{2}\right)^k \frac{k(k+1)(k+2)^2(k+3)(k+4)}{180}$
	$\mathcal{V}_n$	-1	$\frac{1}{2}$	$-\frac{1}{4}$	$\frac{1}{8}$	$-\frac{1}{16}$	$\frac{1}{32}$	$-\frac{1}{64}$	$\frac{1}{128}$	⋯	$-\left(-\frac{1}{2}\right)^{k-1}$
	$\mathcal{N}_n$	0	0	0	0	0	0	0	0	⋯	0
	$\mathcal{H}_n$	4	-1	1	$-\frac{1}{4}$	$\frac{1}{4}$	$-\frac{1}{16}$	$\frac{1}{16}$	$-\frac{1}{64}$	⋯	$\left(-\frac{1}{2}\right)^{k-1}\left[3-(-1)^k\right]$
	$\mathcal{A}_n$	-2	2	$-\frac{3}{2}$	1	$-\frac{5}{8}$	$\frac{3}{8}$	$-\frac{7}{32}$	$\frac{1}{8}$	⋯	$4k\left(-\frac{1}{2}\right)^k$
	$\mathcal{B}_n$	2	$\frac{3}{2}$	$\frac{1}{2}$	$\frac{1}{16}$	0	0	0	0	⋯	0
	$\mathcal{D}_n$	0	0	0	0	0	0	0	0	⋯	0
Harmonic	$\mathcal{T}_n$	-1	4	-9	16	-25	36	-49	64	⋯	$(-1)^k k^2$
	$\mathcal{V}_n$	-1	0	0	0	0	0	0	0	⋯	0
	$\mathcal{N}_n$	0	2	3	-2	3	-2	3	-2	⋯	$\frac{1-(-1)^k 5}{2}$
	$\mathcal{H}_n$	4	-2	4	-2	4	-2	4	-2	⋯	$1-3(-1)^k$
	$\mathcal{A}_n$	1	-2	2	-2	2	-2	2	-2	⋯	$-2(-1)^k$
	$\mathcal{B}_n$	2	1	0	0	0	0	0	0	⋯	0
	$\mathcal{D}_n$	0	1	2	2	2	2	2	2	⋯	2

where $\boldsymbol{r}_i$ denotes the position vector at $\bar{t} = \bar{t}_0$, and $\boldsymbol{\sigma}$ is vector describing the direction of propagation. From the condition (19) we can also see that

$$\left|\frac{d\boldsymbol{x}_0}{d\bar{t}}\right|^2 = |\boldsymbol{\sigma}|^2 = 1 \,. \tag{B.5}$$

This then is the zeroth order solution. For the 1PN equation we have

$$\left(\frac{d^2\boldsymbol{x}_1}{d\bar{t}^2}\right) = \frac{1}{|\boldsymbol{x}|^2}\left[\left(\mathcal{T}_1 + \left|\frac{d\boldsymbol{x}}{d\bar{t}}\right|^2 \mathcal{V}_1 + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)^2 \mathcal{N}_1\right)\frac{\boldsymbol{x}}{|\boldsymbol{x}|} + \left(\frac{\boldsymbol{x}}{|\boldsymbol{x}|}\cdot\frac{d\boldsymbol{x}}{d\bar{t}}\right)\mathcal{H}_1\frac{d\boldsymbol{x}}{d\bar{t}}\right]. \tag{B.6}$$

Because this component of the solution is of order $O(m)$ we can use the zeroth order solution in the right-hand-side to obtain

$$\left(\frac{d^2\boldsymbol{x}_1}{d\bar{t}^2}\right) = \frac{1}{|\boldsymbol{x}_0|^2}\left[\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)^2\right]\frac{\boldsymbol{x}_0}{|\boldsymbol{x}_0|} + \frac{\mathcal{H}_1}{|\boldsymbol{x}_0|^2}\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)\boldsymbol{\sigma}. \tag{B.7}$$

To solve this equation, we separated it into two vectorial components, one along the zeroth order direction of propagation, $\boldsymbol{\sigma}$ and another one along a perpendicular direction. The decomposed form of the first order solution reads

$$\boldsymbol{x}_1 = [\boldsymbol{x}_1 - (\boldsymbol{x}_1\cdot\boldsymbol{\sigma})\,\boldsymbol{\sigma}] + (\boldsymbol{x}_1\cdot\boldsymbol{\sigma})\,\boldsymbol{\sigma} = \boldsymbol{x}_{1\perp} + \boldsymbol{x}_{1\parallel}. \tag{B.8}$$

The perpendicular component lies along the direction of impact parameter vector $\boldsymbol{b}$ defined as

$$\boldsymbol{b} = \boldsymbol{x_0} - (\boldsymbol{x_0}\cdot\boldsymbol{\sigma})\boldsymbol{\sigma} = \boldsymbol{r}_i - (\boldsymbol{r}_i\cdot\boldsymbol{\sigma})\boldsymbol{\sigma} = \boldsymbol{\sigma}\times(\boldsymbol{r}_i\times\boldsymbol{\sigma})\,. \tag{B.9}$$

The decomposition yields two differential equations,

$$\frac{d^2\boldsymbol{x}_{1\parallel}}{d\bar{t}^2} = \left[(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{H}_1)\,\frac{(\boldsymbol{x_0}\cdot\boldsymbol{\sigma})}{|\boldsymbol{x}_0|^3} + \mathcal{N}_1\frac{(\boldsymbol{x_0}\cdot\boldsymbol{\sigma})^3}{|\boldsymbol{x_0}|^5}\right]\boldsymbol{\sigma}, \tag{B.10}$$

$$\frac{d^2\boldsymbol{x}_{1\perp}}{d\bar{t}^2} = \left[\frac{(\mathcal{T}_1 + \mathcal{V}_1)}{|\boldsymbol{x}_0|^3} + \mathcal{N}_1\frac{(\boldsymbol{x_0}\cdot\boldsymbol{\sigma})^2}{|\boldsymbol{x_0}|^5}\right]\boldsymbol{b}, \tag{B.11}$$

that can be directly integrated. The integration requires the evaluation of a particular family of primitives that can be easily found through the recursive relations presented in Appendix F. The resulting first order equations are

$$\frac{d\boldsymbol{x}_{1\parallel}}{d\bar{t}} = \left[-\frac{(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1)}{|\boldsymbol{x}_0|} + \frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{b}|^2}{|\boldsymbol{x_0}|^3}\right]\boldsymbol{\sigma} + \boldsymbol{C}_{\parallel}, \tag{B.12}$$

$$\frac{d\boldsymbol{x}_{1\perp}}{d\bar{t}} = \left[(\mathcal{T}_1 + \mathcal{V}_1)\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right) + \frac{\mathcal{N}_1}{3}\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)^3\right]\frac{\boldsymbol{b}}{|\boldsymbol{b}|^2} + \boldsymbol{C}_{\perp}\,, \tag{B.13}$$

where $\boldsymbol{C}_{\parallel}$ and $\boldsymbol{C}_{\perp}$ are constant integration vectors that must be determined from the boundary conditions of the problem. For this, we once again looked at the null condition. This time, the 1PN form of Eq. (19) implies

$$\frac{\mathcal{A}_1 + \mathcal{B}_1}{|\boldsymbol{x}_0|} + 2\left(\boldsymbol{\sigma}\cdot\frac{d\boldsymbol{x}_1}{d\bar{t}}\right) + \frac{\mathcal{D}_1}{|\boldsymbol{x}_0|}\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)^2 = 0. \tag{B.14}$$

This equation requires the parallel solution's derivative to vanish when $\bar{t}\to\pm\infty$ and thus $\boldsymbol{C}_{\parallel} = 0$. The perpendicular component, conversely, should only vanish before the interaction with the central body, i.e. when $\bar{t}\to-\infty$. This implies that

$$\boldsymbol{C}_{\perp} = \left[\mathcal{T}_1 + \mathcal{V}_1 + \frac{\mathcal{N}_1}{3}\right]\frac{\boldsymbol{b}}{|\boldsymbol{b}|^2}. \tag{B.15}$$

For completeness we write the full solution here

$$\frac{d\boldsymbol{x}_{1\parallel}}{d\bar{t}} = \left[-\frac{(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1)}{|\boldsymbol{x}_0|} + \frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{b}|^2}{|\boldsymbol{x_0}|^3}\right]\boldsymbol{\sigma}, \tag{B.16}$$

$$\frac{d\boldsymbol{x}_{1\perp}}{d\bar{t}} = \left[(\mathcal{T}_1 + \mathcal{V}_1)\left(\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right) + 1\right) + \frac{\mathcal{N}_1}{3}\left(\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)^3 + 1\right)\right]\frac{\boldsymbol{b}}{|\boldsymbol{b}|^2}\,. \tag{B.17}$$

From this two expressions, we obtained the first order solution by direct integration from emission time ($\bar{t}_0$) to an arbitrary time ($\bar{t}$).

$$\boldsymbol{x}_{1\parallel} = \left[-(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1)\log\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma} + |\boldsymbol{x_0}|}{\boldsymbol{r}_i\cdot\boldsymbol{\sigma} + |\boldsymbol{r}_i|}\right) + \frac{\mathcal{N}_1}{3}\left(\frac{(\boldsymbol{x_0}\cdot\boldsymbol{\sigma})}{|\boldsymbol{x_0}|} - \frac{(\boldsymbol{r}_i\cdot\boldsymbol{\sigma})}{|\boldsymbol{r}_i|}\right)\right]\boldsymbol{\sigma}, \tag{B.18}$$

$$\boldsymbol{x}_{1\perp} = \left[\left(\mathcal{T}_1 + \mathcal{V}_1 + \frac{\mathcal{N}_1}{3}\right)\left(\frac{|\boldsymbol{x_0}| - |\boldsymbol{r}_i| + \boldsymbol{\sigma}\cdot(\boldsymbol{x_0} - \boldsymbol{r}_i)}{|\boldsymbol{b}|^2}\right) + \frac{\mathcal{N}_1}{3}\left(\frac{1}{|\boldsymbol{x_0}|} - \frac{1}{|\boldsymbol{r}_i|}\right)\right]\boldsymbol{b}, \tag{B.19}$$

where, again, we made use of the integrals presented in Appendix F. When the metric coefficients associated with the Schwarzschild solution are considered, Eqs. (B.18) and (B.19), match those previously derived in the literature (Klioner & Zschocke 2010; Brumberg 2017).

B.2. Boundary value problem

We then found the trajectory that connects an emitter and observer. This is equivalent to solving the boundary value problem such that $\boldsymbol{x}(\bar{t}_0) = \boldsymbol{r}_i$ and $\boldsymbol{x}(\bar{t}_f) = \boldsymbol{r}_f$. First we noted that, from the Euclidean distance definition,

$$\boldsymbol{D} = \boldsymbol{r}_f - \boldsymbol{r}_i = \boldsymbol{x}(\bar{t}_f) - \boldsymbol{r}_i = \boldsymbol{\sigma}(\bar{t} - \bar{t}_0) + m\boldsymbol{x}_1(\bar{t}_f), \tag{B.20}$$

we could write

$$|\boldsymbol{D}| = \sqrt{\boldsymbol{D}\cdot\boldsymbol{D}} = \sqrt{(\bar{t}_f - \bar{t}_0)^2 + 2m(\boldsymbol{\sigma}\cdot\boldsymbol{x}_1) + m^2|\boldsymbol{x}_1|^2} = (\bar{t}_f - \bar{t}_0) + m(\boldsymbol{\sigma}\cdot\boldsymbol{x}_1) + O(m^2), \tag{B.21}$$

where $\boldsymbol{x}_1 = \boldsymbol{x}_1(\bar{t}_f)$. This expression allowed us to write the original equation as

$$\boldsymbol{D} = \boldsymbol{\sigma}\Big[\,|\boldsymbol{D}| - m(\boldsymbol{\sigma}\cdot\boldsymbol{x}_1)\,\Big] + m\boldsymbol{x}_1 = \boldsymbol{\sigma}|\boldsymbol{D}| + m\Big[\,\boldsymbol{x}_1 - (\boldsymbol{\sigma}\cdot\boldsymbol{x}_1)\,\boldsymbol{\sigma}\,\Big] = \boldsymbol{\sigma}|\boldsymbol{D}| + m\boldsymbol{x}_{1\perp}. \tag{B.22}$$

The last step makes use of the earlier decomposition (B.8). The initial direction of travel is thus, to first order,

$$\boldsymbol{\sigma} = \frac{\boldsymbol{D}}{|\boldsymbol{D}|} - m\frac{\boldsymbol{x}_{1\perp}}{|\boldsymbol{D}|}. \tag{B.23}$$

We could then use this expression to calculate the direction of the photon at the observer's position:

$$\frac{d\boldsymbol{x}}{d\bar{t}} = \frac{d\boldsymbol{x}_0}{d\bar{t}} + m\frac{d\boldsymbol{x}_1}{d\bar{t}} = \boldsymbol{\sigma} + m\frac{d\boldsymbol{x}_1}{d\bar{t}} = \frac{\boldsymbol{D}}{|\boldsymbol{D}|} + m\left(\frac{d\boldsymbol{x}_1}{d\bar{t}} - \frac{\boldsymbol{x}_{1\perp}}{|\boldsymbol{D}|}\right) = \frac{\boldsymbol{D}}{|\boldsymbol{D}|} + m\left(\frac{d\boldsymbol{x}_{1\parallel}}{d\bar{t}}\right) + m\left(\frac{d\boldsymbol{x}_{1\perp}}{d\bar{t}} - \frac{\boldsymbol{x}_{1\perp}}{|\boldsymbol{D}|}\right). \tag{B.24}$$

Furthermore, the first order expression for $\boldsymbol{\sigma}$ in Eq. (B.23), can be used to rewrite the impact parameter, $\boldsymbol{b}$, defined in Eq. (B.9) as

$$\boldsymbol{b} = \boldsymbol{\sigma}\times\left(\boldsymbol{r}_f\times\boldsymbol{\sigma}\right) = \frac{\boldsymbol{D}}{|\boldsymbol{D}|}\times\left(\frac{\boldsymbol{r}_f\times\boldsymbol{D}}{|\boldsymbol{D}|}\right) = -\frac{\boldsymbol{D}\times(\boldsymbol{r}_f\times\boldsymbol{r}_i)}{|\boldsymbol{D}|^2} = \frac{\boldsymbol{D}\times\boldsymbol{L}}{|\boldsymbol{D}|^2} + O(m), \tag{B.25}$$

where we defined the vector $\boldsymbol{L} = \boldsymbol{r}_i\times\boldsymbol{r}_f$. Note that we restricted our calculation to zeroth order in m. This is because any higher order term in Eq. (B.24) will naturally drop out. Note as well, that, from the above expression,

$$|\boldsymbol{b}| = \frac{|\boldsymbol{L}|}{|\boldsymbol{D}|}. \tag{B.26}$$

With this, we could express the perturbation terms in Eq. (B.24) as

$$\left(\frac{d\boldsymbol{x}_{1\parallel}}{d\bar{t}}\right) = -\frac{1}{|\boldsymbol{r}_f|}\left[(\mathcal{T}_1 + \mathcal{V}_1 + \mathcal{N}_1 + \mathcal{H}_1) - \frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{L}|^2}{|\boldsymbol{D}|^2|\boldsymbol{r}_f|^2}\right]\frac{\boldsymbol{D}}{|\boldsymbol{D}|} + O(m) \tag{B.27}$$

$$\left(\frac{d\boldsymbol{x}_{1\perp}}{d\bar{t}}\right) = \left[(\mathcal{T}_1 + \mathcal{V}_1)\left(\left(\frac{\boldsymbol{r}_f\cdot\boldsymbol{\sigma}}{|\boldsymbol{r}_f|}\right) + 1\right) + \frac{\mathcal{N}_1}{3}\left(\left(\frac{\boldsymbol{r}_f\cdot\boldsymbol{\sigma}}{|\boldsymbol{r}_f|}\right)^3 + 1\right)\right]\frac{\boldsymbol{D}\times\boldsymbol{L}}{|\boldsymbol{L}|^2} + O(m) \tag{B.28}$$

$$\left(\frac{\boldsymbol{x}_{1\perp}}{|\boldsymbol{D}|}\right) = \frac{1}{|\boldsymbol{D}|}\left[\left(\mathcal{T}_1 + \mathcal{V}_1 + \frac{\mathcal{N}_1}{3}\right)\Big(\,|\boldsymbol{r}_f| - |\boldsymbol{r}_i| + |\boldsymbol{D}|\,\Big) + \frac{\mathcal{N}_1}{3}\frac{|\boldsymbol{L}|^2}{|\boldsymbol{D}|^2}\left(\frac{1}{|\boldsymbol{r}_f|} - \frac{1}{|\boldsymbol{r}_i|}\right)\right]\frac{\boldsymbol{D}\times\boldsymbol{L}}{|\boldsymbol{L}|^2} + O(m), \tag{B.29}$$

which are the expressions that show up in the deflection formulas presented in Eqs. (26) and (27).

Appendix C: PPN expansion coefficients

Using the PPN metric (48) we can write the expansion coefficients in Eqs. (11), (16), and (19). We write down these coefficients in Table C.1. Note that the connection to the Schwarzschild solution can be directly made by replacing the coefficients by their GR values:

$$\beta = 1, \quad \gamma = 1\,, \quad \eta = \frac{3}{4}, \quad \delta = \frac{3}{8}, \qquad \lambda = \frac{1}{16}. \tag{C.1}$$

Table C.1. Expansion coefficients for the PPN metric.

	n	1	2	3	$\cdots$
PPN metric	$\mathcal{T}_n$	-1	$2(\beta+\gamma)$	$-4\gamma(\beta+\gamma)+4\delta-3\eta$	$\cdots$
	$\mathcal{V}_n$	$-\gamma$	$2(\gamma^2-2\delta)$	$-4(\gamma^3-3\gamma\delta+3\lambda)$	$\cdots$
	$\mathcal{N}_n$	0	0	0	$\cdots$
	$\mathcal{H}_n$	$2(1+\gamma)$	$4(1-\beta-\gamma^2+2\delta)$	$8-12\beta+8\gamma^3-24\gamma\delta+6\eta+24\lambda$	$\cdots$
	$\mathcal{A}_n$	-2	2β	-2η	$\cdots$
	$\mathcal{B}_n$	2γ	4δ	8λ	$\cdots$
	$\mathcal{D}_n$	0	0	0	$\cdots$

Table D.1. Expansion coefficients of the Reissner-Nordström metric in standard coordinates.

	n	1	2	3	4	5	6	7	$\cdots$	k
Reissner-Nordström	$\mathcal{T}_n$	-1	$2+\Theta^2$	$-3\Theta^2$	Θ^4	0	0	0	$\cdots$	0
	$\mathcal{V}_n$	-2	Θ^2	0	0	0	0	0	$\cdots$	0
	$\mathcal{N}_n$	3	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	$\cdots$	θ_k
	$\mathcal{H}_n$	$2\theta_1$	$2\theta_2$	$2\theta_3$	$2\theta_4$	$2\theta_5$	$2\theta_6$	$2\theta_7$	$\cdots$	$2\theta_k$
	$\mathcal{A}_n$	-2	Θ^2	0	0	0	0	0	$\cdots$	0
	$\mathcal{B}_n$	0	0	0	0	0	0	0	$\cdots$	0
	$\mathcal{D}_n$	η_1	η_2	η_3	η_4	η_5	η_6	η_7	$\cdots$	η_k

Appendix D: Reissner-Nordström solution

The Reissner–Nordström solution represents the gravitational field produced by a charged, non-rotating, spherically symmetric body (Reissner 1916; Nordström 1918). When written in standard coordinates, the metric takes the form presented in Eq. (52). When written in terms of the dimensionless charge parameter $\Theta^2 = q^2/m^2$, it reads

$$p(r) = \left(1 - \frac{2m}{r} + \frac{2\Theta^2 m^2}{r^2}\right), \qquad q(r) = \frac{1}{p(r)}, \qquad a(r) = r. \tag{D.1}$$

This form of the metric allows simple analytical expression for the Cartesian metric functions defined in Eq. (7). The expansion coefficients are presented in Table D.1 where we define the helper quantities

$$\theta_k = \frac{\left(1+\sqrt{1-\Theta^2}\right)^k + \left(1-\sqrt{1-\Theta^2}\right)^k}{2}$$
$$\eta_k = \frac{\left(1+\sqrt{1-\Theta^2}\right)^{k+1} - \left(1-\sqrt{1-\Theta^2}\right)^{k+1}}{2\sqrt{1-\Theta^2}}. \tag{D.2}$$

Notice how the expansion coefficients of the Schwarzschild solution in standard coordinates presented in Table A.1 are retrieved when $\Theta \to 0$.

Appendix E: Transformation to isotropic coordinates

The isotropic form of a spherically symmetric space-time forces the equations of motion to take a simpler form since the cross metric term $D(r)$ in Eq. (3) is identically zero. We show how any spherically symmetric space-time of the form (2) can be brought into this form. Given a space-time, whose line element reads

$$ds^2 = -p(r)c^2dt^2 + q(r)dr^2 + a(r)^2d\Omega^2, \tag{E.1}$$

we wish to convert the radial coordinate such that the line element becomes

$$ds^2 = -P(\rho)c^2dt^2 + Q(\rho)\left[d\rho^2 + \rho^2 d\Omega^2\right], \tag{E.2}$$

where ρ represents the new isotropic radial coordinate. Matching the coefficients of dt^2 and $d\Omega^2$ implies

$$P(\rho) = p(r), \quad a^2(r) = \rho^2 Q(\rho), \tag{E.3}$$

while the relation between the radial differentials implies $Q(\rho)d\rho^2 = q(r)dr^2$. Together with the above relations, we obtain the integral equation

$$\int \frac{d\rho}{\rho} = \pm \int \frac{\sqrt{q(r)}}{a(r)} dr, \tag{E.4}$$

where the sign is chosen such that ρ increases with r. Once the integral on the right is solved, a relation between ρ and r is obtained and relations (E.3) can be used to write $P(\rho)$ and $Q(\rho)$. We apply this procedure to the Reissner-Nordström metric (52). The relation between the standard and isotropic coordinate turns out to be

$$\rho = \frac{r - m + \sqrt{r^2 - 2mr + 2q^2}}{2}, \tag{E.5}$$

or inverting the relation for r,

$$r = \rho\left(1 + \frac{m}{2\rho}\right)^2 - \frac{q^2}{2\rho}, \tag{E.6}$$

which reduces to the Schwarzschild isotropic relation (A.2) when the charge parameter goes to zero. The isotropic functions read

$$P(\rho) = \frac{\left(1 - \frac{m^2}{4\rho^2} + \frac{q^2}{2\rho^2}\right)^2}{\left(\left(1 + \frac{m}{2\rho}\right)^2 - \frac{q^2}{2\rho^2}\right)^2}, \quad Q(\rho) = \left(\left(1 + \frac{m}{2\rho}\right)^2 - \frac{q^2}{2\rho^2}\right)^2. \tag{E.7}$$

Appendix F: Useful Integrals and relations

Analytically solving the PN equations of motion requires evaluating integrals of the form

$$\mathbb{Q}_q^k = \int \frac{1}{|\boldsymbol{x}_0|^q}\left(\frac{\boldsymbol{x_0}\cdot\boldsymbol{\sigma}}{|\boldsymbol{x_0}|}\right)^k d\bar{t}, \tag{F.1}$$

where q and k are positive integers and

$$\boldsymbol{x}_0(\bar{t}) = \boldsymbol{r}_i + \boldsymbol{\sigma}(\bar{t} - \bar{t}_0). \tag{F.2}$$

Defining $f(t) = |\boldsymbol{x_0}|$, the integral can be seen to take the general form

$$\mathbb{Q}_q^k = \int \frac{\dot{f}^k}{f^q} d\bar{t} = \int \frac{\dot{f}^{k-1}}{f^q} df, \tag{F.3}$$

where $\dot{f}$ denotes the derivative with respect to $\bar{t}$. A further simplification can be done by making use of the impact parameter definition of Eq. (B.9). Since

$$b^2 = |\boldsymbol{x}_0|^2 - (\boldsymbol{x}_0\cdot\boldsymbol{\sigma})^2 = f^2\left(1 - \dot{f}^2\right), \tag{F.4}$$

integral (F.3) can be written solely in terms of f as

$$\mathbb{Q}_q^k = \int \frac{1}{f^q}\left(1 - \frac{b^2}{f^2}\right)^{\frac{k-1}{2}} df\ , \tag{F.5}$$

or, in terms of the variable $w = b/f$, as

$$\mathbb{Q}_q^k = -\frac{1}{b^{q-1}} \int w^{q-2}\left(1 - w^2\right)^{\frac{k-1}{2}} dw. \tag{F.6}$$

We show that this class of integrals can be solved analytically and satisfy particular recursive relations.

F.1. Case $q \geq 2$

When $q \geq 2$, the integral can be solved using trigonometric substitutions. Let n be a positive integer. Substituting $w = \cos(u)$ or $w = \sin(u)$, the following recursive relations can be found for the integrals

$$\mathbb{Q}^k_{n+2} = \frac{1}{b^{n+1}}\left[\frac{w^{n-1}(1-w^2)^{\frac{k+1}{2}}}{n+k}\right] + \left(\frac{n-1}{n+k}\right)\frac{\mathbb{Q}^k_n}{b^2}$$

$$\mathbb{Q}^k_{n+2} = \frac{1}{b^{n+1}}\left[\frac{w^{n+1}(1-w^2)^{\frac{k-1}{2}}}{k+n}\right] + \left(\frac{k-1}{k+n}\right)\mathbb{Q}^{k-2}_{n+2}.$$

The first relation maps the value (k, q) to $(k, q+2)$ while the second relation maps (k, q) to $(k+2, q)$. This means that, to evaluate any integral with $q > 2$ ($n > 0$) we only need to have evaluate $\mathbb{Q}^0_2$, $\mathbb{Q}^1_2$, $\mathbb{Q}^0_3$, and $\mathbb{Q}^1_3$ since all other values can be obtained by applying the recursive relations. In terms of w, the required integrals read

$$\begin{aligned} \mathbb{Q}^0_2 &= -\frac{\arcsin(w)}{b} \qquad & \mathbb{Q}^1_2 &= -\frac{w}{b} \\ \mathbb{Q}^0_3 &= \frac{\sqrt{1-w^2}}{b^2} \qquad & \mathbb{Q}^1_3 &= -\frac{w^2}{2b^2}. \end{aligned} \tag{F.7}$$

Besides these recursive formulas, the integral also has symmetry relations. If we explicitly write the integration variable as $\mathbb{Q}^k_{n+2}[w]$, the integrals can be seen to have the symmetry

$$\mathbb{Q}^k_{n+2}[w] = -b^{(k-n)}\mathbb{Q}^n_{k+2}\left[\sqrt{1-w^2}\right], \tag{F.8}$$

which also holds for negative values of n. Notice how this relation holds between $\mathbb{Q}^1_2$ and $\mathbb{Q}^0_3$ in Eq. (F.7).

F.2. Case $q = 1$

Using the substitution $w = \sqrt{1-u^2}$ and completing the square, the following relation is obtained

$$\mathbb{Q}^k_1 = \mathbb{Q}^{k-2}_1 - \frac{1}{k-1}\left(1-w^2\right)^{\frac{k-1}{2}}. \tag{F.9}$$

Only two integrals are required to use the recursive relations These read

$$\mathbb{Q}^0_1 = \operatorname{arctanh}\left(\sqrt{1-w^2}\right), \qquad \mathbb{Q}^1_1 = -\log(w). \tag{F.10}$$

F.3. Case $q = 0$

Using the same trigonometric substitution as for the $q = 1$ case, the following recursive relation is found:

$$\mathbb{Q}^k_0 = \mathbb{Q}^{k-2}_0 - b^2\mathbb{Q}^k_2. \tag{F.11}$$

The base integrals are $\mathbb{Q}^0_2$ and $\mathbb{Q}^1_2$ written in Eq. (F.7) plus the $\mathbb{Q}^0_0$ and $\mathbb{Q}^1_0$ given by

$$\mathbb{Q}^0_0 = b\frac{\sqrt{1-w^2}}{w}, \qquad \mathbb{Q}^1_0 = \frac{b}{w}. \tag{F.12}$$

The above integrals simplify to $\mathbb{Q}^0_0 = \bar{t}$ and $\mathbb{Q}^1_0 = f$, respectively.

F.4. Special case: Odd k or q

When k is odd, the integrals can be evaluate without the need for the recursive formulae. For $k = 1$ direct integration yields

$$\mathbb{Q}^1_q = \begin{cases} \log(f) & for q = 1 \\ \frac{1}{(1-q)f^{q-1}} & \text{for } q \neq 1 \end{cases}. \tag{F.13}$$

For the generic case where $k = 2n+1$, direct use of the binomial expansion formula and power integrations rules yields

$$\mathbb{Q}^{2n+1}_q = \mathbb{Q}^1_q + \left(\frac{1}{f}\right)^{q-1}\sum_{j=1}^{n}\frac{(-1)^{j+1}}{(q-1)+2j}\left(\frac{b}{f}\right)^{2j}. \tag{F.14}$$

Note that because of the symmetry of the integrals, these relations are easily translated into expressions for odd values of q.